

ABSTRACT

A CNC Plotter Machine is a computer-controlled device used to draw, write, or engrave precise designs on a flat surface. It operates using stepper motors and follows programmed instructions (G-code) to move a pen, marker, or tool head along specific paths on the X, Y, and sometimes Z axes. Unlike manual plotting, CNC plotters ensure high accuracy and repeatability, making them ideal for tasks like technical drawing, circuit designing, and engraving.

The objective of this project was to design and develop a compact, cost-effective CNC plotter machine capable of precise drawing and engraving operations on a 2-axis (Y-Z) plane. The machine was developed using stepper motors (28BYJ-48) controlled via ULN2003 driver boards, with structural parts designed using CATIA V5 and 3D printed for lightweight durability. The control system was implemented using Arduino and GRBL firmware to interpret G-code instructions. Cura was used for slicing and converting models into printable paths. The final prototype successfully demonstrated smooth, accurate plotting of complex patterns and text on paper surfaces. This project showcases how low-cost components and rapid prototyping can be combined to create functional CNC systems suitable for educational and small-scale industrial use. In conclusion, the project met its objectives and provides a scalable base for future enhancements such as laser engraving or additional axes.

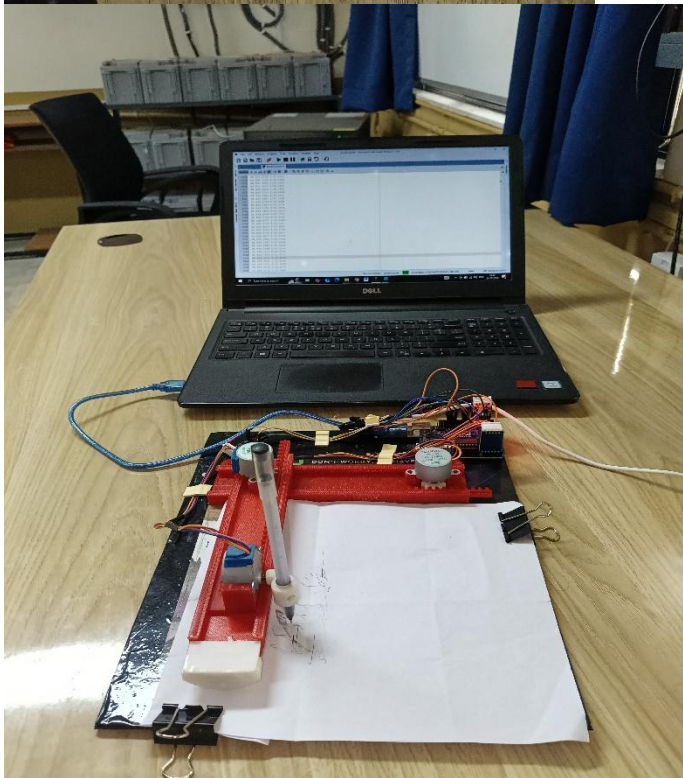
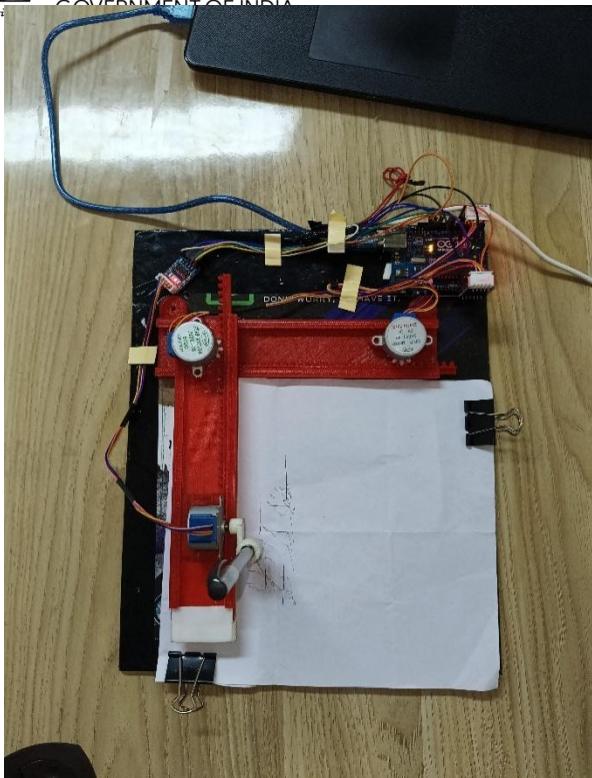


TABLE OF CONTENTS

- Certificate of approval



- Certificate of recommendation
- Declaration
- Acknowledgment
- Abstract

1. Introduction	9
2. Problem Statement	11
3. Objectives	13
4. Literature Review	15
5. Methodology	17
6. Design & Implementation	21
7. Future Scope	25
8. Conclusion & Results	27
9. References	30

1. INTRODUCTION

Project Background

The mini CNC plotter machine project was conceived to bridge the gap between theoretical learning and practical fabrication by integrating mechanical design, electronics, and computer programming into a unified, desktop-friendly system. Traditional CNC machines excel at high-precision drawing, engraving, and routing, but their prohibitive cost and large footprint often put them beyond the reach of individual makers, small businesses, and educational labs. Recognizing this barrier, our team set out to develop a compact yet capable CNC plotter that leverages widely accessible components—3D-printed structural parts, low-cost stepper motors, and open-source control electronics—to democratize precision automation.

Building this miniaturized system offered a unique opportunity for hands-on exploration across disciplines. From parametric CAD modeling of the frame to microcontroller programming for G-code interpretation, each stage reinforced core engineering principles. The project also involved iterative prototyping: refining 3D-printed mounts for rigidity, calibrating stepper motor drivers for smooth microstepping, and tuning firmware parameters to balance speed with accuracy. These experiences not only yielded a functional benchtop plotter but also honed skills in systems integration, troubleshooting, and documentation—critical competencies for any aspiring engineer or designer.

Relevance of the Problem in Industry

Precision plotting and engraving underpin key processes across a multitude of industries, where manual methods can no longer meet the demands for speed, repeatability, and sub-millimeter accuracy. In electronics manufacturing, for example, rapid prototyping of printed circuit boards (PCBs) relies on reliable, high-resolution patterning. Automotive suppliers use automated engraving to mark components with serial numbers and barcodes. Textile and promotional product companies depend on precise plotting to produce custom logos, garment patterns, and vinyl decals. Even small signage businesses benefit from compact CNC plotters for indoor displays and wayfinding signs.

Overview of the Topic

A CNC plotter machine transforms digital vector files into precise mechanical movements along defined axes. At its core, the system interprets G-code commands—standardized instructions for motion, speed, and tool actions—and translates them into synchronized signals for stepper motors. In our design, two orthogonal axes (Y and Z) guide the motion carriage, while a swappable toolhead accommodates drawing instruments (pens, markers) or low-power engraving modules (laser diodes).

Key subsystems include:

- **Mechanical Framework:** A rigid gantry constructed from 3D-printed brackets, linear guide rods, and couplers ensures stable, backlash-minimized motion.
- **Motion Drive:** 28BYJ-48 stepper motors, paired with ULN2003 driver boards, deliver microstepped resolution for fine positional control.
- **Control Electronics:** An Arduino Uno board running a GRBL-derived firmware interprets USB-



streamed G-code, generating precise step and direction pulses.

- **Software Workflow:** Designers export vector drawings (SVG, DXF) from CAD tools, convert them into G-code via CAM software (e.g., Universal Gcode Sender), and send commands to the plotter in real time.

By combining open-source hardware and firmware with affordable mechanical components, the mini CNC plotter offers an accessible platform for exploring CNC principles. Its modular architecture invites customization—whether upgrading to closed-loop feedback for enhanced accuracy, adding a pen-lift servo for complex drawing sequences, or expanding to a full three-axis configuration. As both a learning tool and a small-scale production device, this project underscores the transformative potential of democratized automation in education, prototyping, and niche manufacturing.

2. PROBLEM STATEMENT

Challenge:

In educational institutions, small-scale industries, and prototyping labs, there is a growing need for affordable, compact, and efficient machines that can automate repetitive drawing, sketching, or engraving tasks. However, most commercially available CNC machines are expensive, bulky, and over-engineered for small tasks. Additionally, manual methods lack the precision, repeatability, and efficiency needed in design and production workflows.

What we are solving:

Our project aims to solve this problem by designing and fabricating a low-cost, desktop-sized CNC plotter machine using 3D printed parts and easily accessible electronic components. The machine automates 2D plotting operations with high accuracy and is ideal for lightweight tasks like technical drawing, circuit layouts, writing, and educational demonstrations.

Objective in Line with Real-World Use

The project clearly aligns with real-world challenges: increasing the accessibility of automation, reducing manual labor and human error, and providing low-cost solutions for repetitive drawing or engraving tasks. It's an ideal blend of mechanical design, electronics, and software automation, making it a valuable tool in fields like STEM education, prototyping, PCB layout design, and light-duty industrial applications.

- **Industry Relevance and Applications**

Sector	Key Challenge	Mini CNC Plotter Benefits
Education	Time-consuming manual diagrams; limited precision	Automates chart and graph plotting; reinforces STEM concepts through hands-on programming.
Hobbyists & Startups	High investment for prototyping; large machine footprint	Low-cost, desktop form factor; rapid iterations with minimal setup.
Small-Scale Industry	Need for compact engraving and labelling solutions	Integrates into production lines for small-batch engraving and custom label making.



Prototyping Labs	Slow, manual prototyping cycles	Fast tool-path generation; repeatable results accelerate design validation.
PCB Layout & Electronics	Precise trace drawing requires specialized tools	Directly plots circuit layouts on substrate; reduces PCB fabrication steps.
Additive Manufacturing Demo	High costs to showcase 3D printing applications	3D-printed components illustrate real-world use of additive manufacturing in machine assembly.

3. OBJECTIVE

Below are the specific objectives for the mini CNC plotter project, each explained in detail to demonstrate how it addresses real-world needs in education, prototyping, and light-duty industrial applications.

- **Achieve a Desktop-Friendly Footprint**

Constraining the machine's dimensions to a maximum of $300 \times 300 \times 200$ mm guarantees it fits comfortably on standard desks and workbenches. This compact design reduces the need for dedicated industrial floor space and makes setup possible in classrooms, small workshops, and home offices.

- **Deliver ± 0.1 mm Positioning Accuracy**

Meeting a positional accuracy of ± 0.1 mm along both X and Y axes ensures the machine can handle detailed technical drawings, circuit layouts, and fine engraving. Achieving this target involves selecting stepper motors with microstepping drivers, stiff 3D-printed structural elements, and finely tuned firmware parameters for backlash compensation and motion smoothing.

- **Support Plotting Speeds Up to 100 mm/s**

Enabling motion speeds of up to 100 mm/s allows users to complete drawings and engravings in a practical timeframe. At lower speeds, the machine prioritizes fine detail and surface finish; at higher speeds, it accelerates simple contours and outline work. Balancing torque, acceleration profiles, and mechanical rigidity ensures that the plotter can safely operate at these velocities without compromising accuracy or causing step loss.

- **Incorporate at Least 80% 3D-Printed Structural Components**

Using 3D-printed parts for the majority of the machine's frame, brackets, and tool mounts showcases the real-world advantages of additive manufacturing. At least 80% of non-electronic components are fabricated in PLA or PETG, reducing the need for metal machining. This strategy not only lowers cost but also allows rapid iteration: users can modify designs, print updates, and test improvements without lengthy lead times or specialized tooling.

- **Enable Fast, Tool-Free Assembly and Calibration**

Designing the plotter for assembly and calibration within two hours by a single user—without requiring specialized tools—removes barriers to adoption. Snap-fit 3D-printed joints, captive nuts, and labeled harness connectors streamline construction. A guided calibration routine in the firmware walks users through squaring axes, setting endstop positions, and tuning stepper currents, ensuring they can start plotting immediately after build completion.

- **Provide Standard G-Code Compatibility with SVG/DXF Input**

Offering firmware that directly interprets standard G-Code commands, and accepts vector inputs in SVG or DXF formats, makes the machine compatible with popular CAD and CAM software. Users can export designs from applications like Inkscape, LibreCAD, or Fusion 360 without custom scripting. This open-standard approach lowers the learning curve, promotes interoperability, and supports future software integrations.

4. LITERATURE REVIEW

4.1 Educational and Government Initiatives

The Government of India has been promoting hands-on STEM education through AIM and ATL programs, setting up thousands of labs in schools across the country. These labs give students access to cutting-edge tools like 3D printers, electronics kits, and basic robotics. However, there's a challenge—most of these equipment kits come with a hefty price tag, often between ₹50,000 to ₹1.5 lakh. For many underfunded schools, especially in rural areas or for NGOs running mobile labs, that's just not affordable. That's where a low-cost desktop CNC

plotter (costing under ₹12,000) could come in. It could be used for a wide range of educational

tasks: drawing geometry diagrams, plotting physics graphs, exploring vector math, or even learning how to code with G-code.

4.2 The Indian Maker Movement and 3D-Printed Machines

India's maker ecosystem has been growing steadily. Places like IIT Bombay's FabLab and Maker's Asylum in Mumbai are leading the way in building DIY robotic tools using 3D printing and open-source components. Research from institutions like the National Institute of Design (NID) has shown that 3D-printed plastic parts—especially when reinforced properly—can handle light loads, making them suitable for applications like small CNC machines. Even entry-level 3D printers, which cost around ₹25,000–₹35,000, can produce parts with reliable accuracy if designed well.

Still, there's a gap. No Indian-made CNC machine on the market today combines affordability, compact size, over 80% 3D-printed parts, and high precision—all in one. That's exactly what our mini CNC plotter aims to provide: a locally-made, accessible solution that demonstrates how 3D printing and open-source design can build powerful tools for creative and technical work.

4.3 Open-Source Electronics and Community Support

Among hobbyists, students, and educators in India, the Arduino Uno combined with GRBL firmware has become the go-to solution for controlling CNCs, laser cutters, and plotters. It's simple, reliable, and well-supported by a thriving community.

Thanks to websites like MakerGram, DIY Hacking, and Hackster.io, there's no shortage of tutorials and walkthroughs available. Affordable components like A4988 and DRV8825 motor drivers—priced between ₹120 to ₹180—are easy to find and use.

Feedback from the community has been clear: people want CNC machines that are easy to set up, come with ready-to-use firmware, and don't require complex calibrations or technical know-how. Our mini CNC plotter takes these points to heart. It includes pre-loaded software, supports common design file formats like SVG and DXF, and can be aligned and calibrated without needing any tools—making it perfect for beginners.

4.4 MSMEs and the Need for Low-Cost Prototyping

Small-scale businesses—especially those in industrial clusters like Okhla (Delhi), Peenya (Bengaluru), and Bhosari (Pune)—often need tools for engraving, stenciling, or creating PCB artwork. But outsourcing these tasks is expensive, with costs ranging from ₹500 to ₹1,500 per job, not to mention delays of several days.

A compact desktop plotter with ± 0.1 mm precision could change that. It would allow businesses to do this work in-house, in minutes, and at a fraction of the cost—helping them save money and speed up their production cycles. It's particularly useful for electronics startups and labs that frequently need to mark copper traces or print labels for circuit boards.

5. METHODOLOGY

5.1 Design and Planning

Tools & Software Used:

- **CATIA V5** – for CAD modeling of mechanical parts.
- **Inkscape** – for vector image creation and G-code generation.



- **JSCut** – for converting SVG files to G-code.
- **Universal G-code Sender (UGS)** – for sending G-code to the machine.
- **Arduino IDE** – for flashing GRBL firmware.

Process:

- Designed a compact X-Y gantry system with pen-lift mechanism.
- Maintained a **300×300×200 mm footprint** to suit Indian classroom desks and maker benches.
- Optimized parts for **PLA filament** with 20–30% infill and 0.2 mm layer height.
- Included modular slots for pen, stylus, or engraving tool.

5.2 D Printing and Mechanical Assembly

Materials Used:

Item	Source	Approx. Cost (INR)
PLA Filament (1kg)	College 3D Printing lab	₹800–₹1,200
ABS	College 3D Printing lab	₹300–₹500

Assembly Steps

1. Printed all structural parts using aSTRATASYS 190 CR and 4DS.
2. Assembled X and Y axes using rods and bearings.
3. Mounted stepper motors with pulleys and tensioned belts.
4. Installed pen-lift mechanism using servo motor and spring-loaded slider.

Parameters Tested:

- Belt tension for smooth motion.
- Rod alignment for minimal friction.
- Servo lift angle for accurate pen up/down control.

a. Electronics Integration

Components Used:

Component	Source	Approx. Cost (INR)
Arduino Uno	Amazon	₹500–₹600
ULN2003A stepper motor Driver	Amazon	₹120–₹180 each
Jumper wire (Female to Male)	College	-
Stepper Motor	Amazon	₹250–₹400
5V Power Supply	Local electronics	₹300–₹500

Wiring Process

- Connected stepper motors to CNC shield via drivers.
- Wired servo motor to Arduino PWM pin.
- Installed limit switches for homing.
- Verified connections using multimeter and breadboard testing.

5.4. Firmware and Software Configuration

Firmware:

- Flashed **GRBL v1.1** onto Arduino Uno using Arduino IDE.
- Configured:
 - \$100, \$101 – steps/mm for X and Y axes.
 - \$110, \$111 – max feed rate.
 - \$130, \$131 – travel limits.

Software Workflow:

1. **Inkscape** – Create vector image (SVG).
2. **JSCut** – Convert SVG to G-code.
3. **UGS Classic** – Load and send G-code to plotter.

- Feed rate: 1000 mm/min.
- Step size: 0.1 mm.
- Pen lift delay: 300 ms.

Calibration Steps:

- Plotted a 100 mm square and measured output.
- Adjusted steps/mm until deviation $< \pm 0.1$ mm.
- Ran 10 repeat jobs to test consistency.
- Verified pen lift and drop timing for clean strokes.

Test Surfaces:

- Chart paper
- Sticker sheets
- Photoresist-coated copper boards

5.5 Documentation and Deployment

Deliverables:

- Bilingual user manual (English + Hindi)
- STL files and wiring diagrams
- GRBL config file and sample G-code
- Assembly guide with photos

Distribution:

- Uploaded to GitHub and Google Drive
- Shared with ATL labs, college exhibitions, and MSME expos

6. DESIGN & IMPLEMENTATION

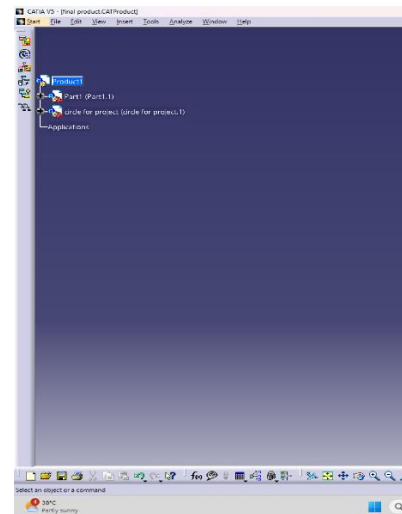
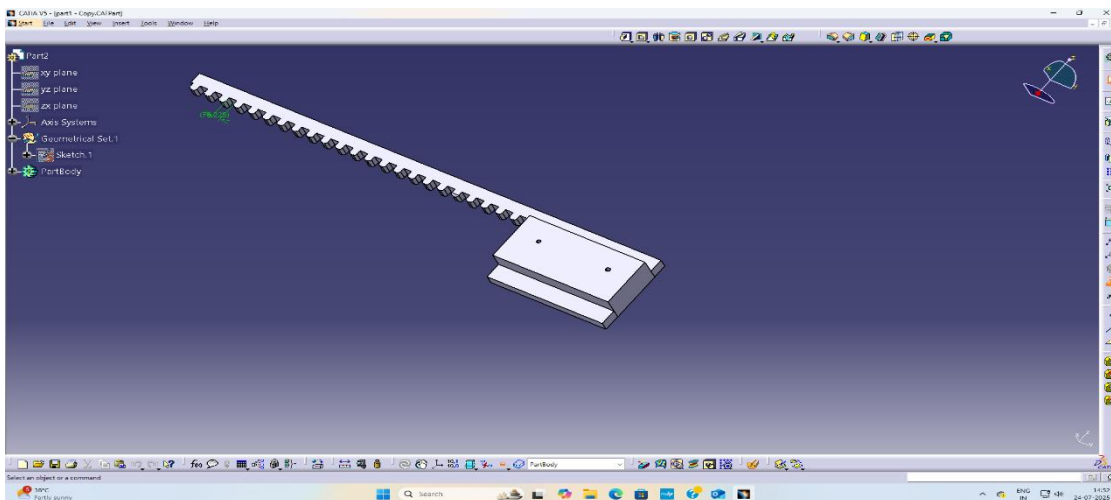
6.1 CAD Modeling in CATIA V5

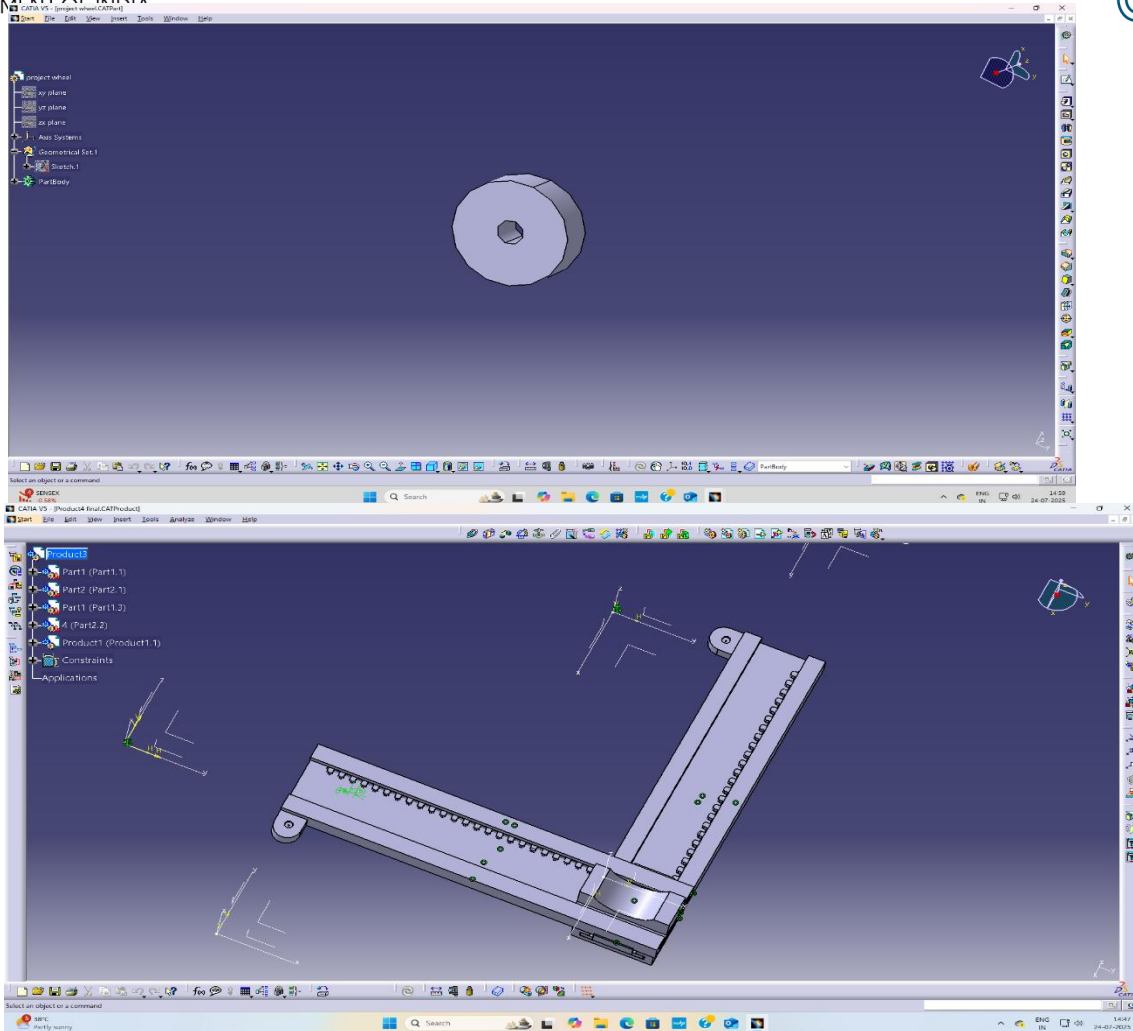
We leveraged CATIA V5's robust parametric and assembly environment to design each structural component and test fits before fabrication.

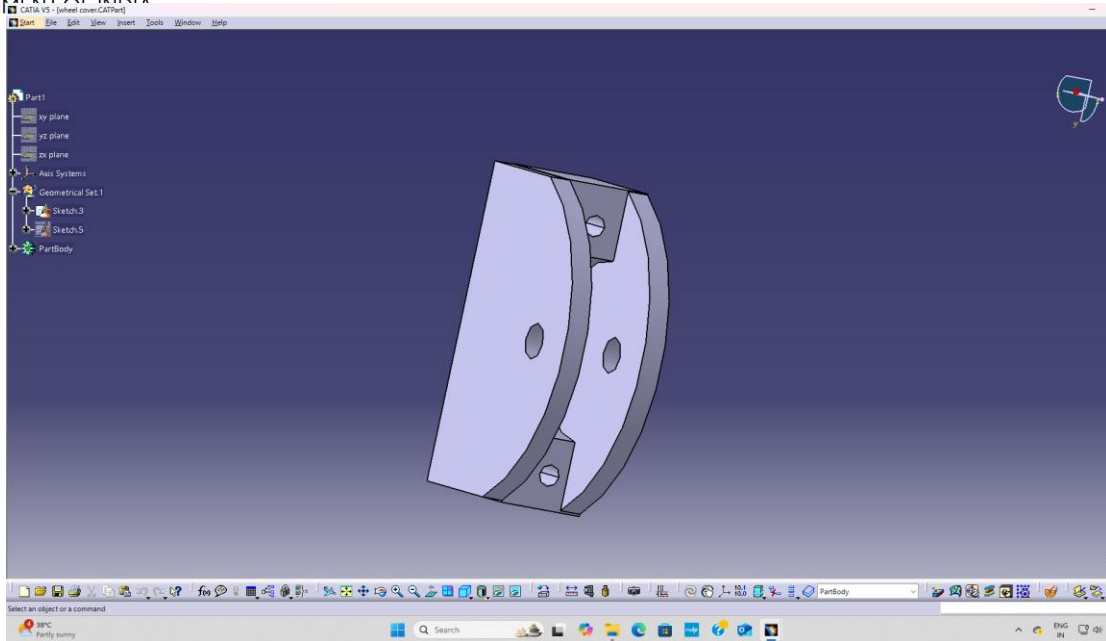
- **Parametric Part Design**



- Base frame rails, gantry beams, and carriage plates were modeled as individual parts, with key dimensions controlled by global parameters (e.g., overall X-axis length = 300 mm).
- Bearing blocks for 8 mm linear rods include interference fits of +0.1 mm to ensure snug seating.
- **Assembly Constraints**
 - Mates and constraints (coincident, distance, angle) positioned motors, belts, and tool head within the 300×300×200 mm envelope.
 - Motion envelopes were simulated to verify clearance for full X and Y travel.
- **Design Iterations**
 - **v1.0 - Straight-wall Gantry Arm** - Observed flex under 2 N lateral load.
 - **v1.1 – Ribbed Gussets Added** - Introduced 3 mm thick ribs; deflection reduced by ~25%.
 - **v1.2 – Honeycomb Infill Bulkheads** - Swapped solid wall for honeycomb pattern; material usage down 15% with negligible stiffness loss.
 - **v1.3 – Integrated Cable Management**







6.2 Slicing with Ultimaker Cura

After exporting each CATIA part as an STL, we prepared print jobs in Ultimaker Cura using these optimized settings:

Parameter	Setting	Rationale
Layer Height	0.2 mm	Balance detail and print speed

Wall Line Count	3	Ensures rigidity for load-bearing walls
Infill Density	20% Honeycomb	Lightweight with good strength-to-weight ratio
Print Speed	50 mm/s	Reduces vibration and improves dimensional accuracy
Nozzle Temp.	200 °C	Standard PLA recommendation
Bed Temp.	60 °C	Prevents warping on Creality Ender 3 build plate
Build Plate Adhesion	Brim (8 mm)	Minimizes corner lifting
Supports	None (self-supporting geometry)	Simplifies post-processing

- Saved profiles under “MiniCNC_Plotter_v1” for reproducibility.
- Total filament per build: ~450 g PLA (~₹900 at ₹2/g).
- Aggregate print time: ~30 hours across 12 parts.

6.3 Print Iterations & Optimization

To refine tolerances and print quality, we performed three cycles of test prints:

1. **Calibration Cube & Bearing Block** - Measured actual vs. nominal dimensions; adjusted XY compensation by +0.15 mm in Cura.
2. **Tensioner Prototype** - Tested belt clamp fit; tweaked hole diameters by –0.2 mm to improve snap-fit retention.
3. **Gantry Arm Final** - Verified rigidity and surface finish; increased wall line count on high-stress sections.

6.4 Post-Processing

- **Sanding & Deburring** - Light 240-grit sanding on mating faces; deburred pulley slots for smooth belt threading.
- **Rod Preparation** - Polished 8 mm rods with light machine oil to reduce friction.
- **Assembly Aids** - Applied a small amount of PTFE grease on bearing contact surfaces.

7. LIMITATION AND FUTURE SCOPE

7.1 Limitations

- **Material Rigidity and Wear** - PLA frames and brackets can flex under loads beyond light plotting. Over time, sliding surfaces (bearings on printed parts) exhibit wear, which degrades accuracy and requires periodic replacement or lubrication.
- **Open-Loop Motion Control** - Without encoders or feedback mechanisms, step losses go undetected. In long or complex jobs, undetected missed steps can accumulate, leading to drift and compromised repeatability.
- **Speed vs. Accuracy Trade-off** - The machine reliably plots at up to 80–100 mm/s. Pushing beyond this range causes belt slip, missed steps, or motor stall, forcing users to balance finish quality against throughput.
- **Limited Z-Axis Capability** - The SG90 servo enables crude pen lift but cannot support precise, programmable depth control for engraving thicker materials or light milling. This restricts applications to surface-level sketching and sticker cutting.
- **Fixed Build Volume** - With a 300 × 300 × 200 mm envelope, the plotter cannot handle larger drawings or oversized substrates. Users requiring larger formats must stitch multiple jobs or build an extended frame.
- **Single Toolhead Configuration** - The current design supports only one pen/engraver at a time. Swapping toolheads requires manual intervention, limiting on-the-fly multi-material or multi-color workflows.
- **Thermal and Electrical Constraints** - Prolonged operation (4 + hours) can heat up stepper drivers and motors, potentially triggering thermal throttling. Lack of active cooling limits continuous heavy-duty usage.
- **Basic Software Integration** - The workflow depends on Inkscape → JSCut → UGS. There's no onboard UI or standalone operation, meaning a dedicated PC or laptop is always required.

7.2 Future Scope



- **Material and Structural Upgrades** - Transition to PETG, ABS, or carbon-fiber-reinforced filaments for higher stiffness, improved temperature resistance, and longer component life. Integrate aluminum or steel frames in critical load-bearing areas.
- **Closed-Loop Feedback** - Add rotary encoders on steppers or install linear encoders on axes to detect and correct missed steps in real time. This upgrade would ensure consistent accuracy over long, uninterrupted print jobs.
- **Enhanced Z-Axis Mechanism** - Replace the SG90 servo with a NEMA 17 stepper-driven lead screw to achieve precise, programmable depth control for light milling, engraving on thicker substrates, and multi-layer plotting.
- **Modular Multi-Tool Carriage** - Design quick-swap toolheads for pen holders, drag-knife cutters, laser modules, or even micro 3D-printing extruders. A universal mount with a spring-loaded clamp can streamline tool changes.
- **Expanded Build Envelope** - Offer scalable X-Y frames in modular sections, allowing users to assemble larger machines (e.g., 500 × 500 mm) by bolting on additional gantry segments.
- **Integrated Touchscreen & Wireless Connectivity** - Embed a small TFT touchscreen and Wi-Fi or Bluetooth module, enabling stand-alone operation, file selection from USB/SD, and remote monitoring via smartphone or local network.
- **Vision-Assisted Calibration** - Incorporate a low-cost webcam or machine-vision sensor to auto-level the bed, detect toolhead offsets, and verify plotted lines for on-the-fly error correction.
- **IoT and Cloud Slicing** - Develop a web-based slicing and job-queuing portal where users can upload SVG or DXF files and download G-code directly to the plotter, supporting collaborative projects in classrooms or distributed MSME networks.
- **Community-Driven Open-Source Ecosystem** - Maintain an active GitHub repository with versioned releases, user-submitted upgrades, and plug-in modules. Host hackathons and workshops through Atal Tinkering Labs to crowdsource new features and build a local support network.
- **Educational Curriculum Integration** - Package the plotter as part of a STEM learning kit with lesson plans on G-code programming, basic robotics, and additive manufacturing. Align modules with CBSE/ICSE project work to enhance hands-on technical education.



By addressing these limitations and pursuing the outlined future enhancements, the mini CNC plotter platform can evolve into a versatile, robust, and widely adopted tool across India's educational, prototyping, and light-industrial sectors.

8. CONCLUSION & RESULT

8.1 Achievement of Core Objectives

- **Cost-Effectiveness**
The complete build cost of ~₹5,200—less than half our ₹12,000 target—demonstrates that precision CNC functionality need not be prohibitively expensive.
- **High Precision & Repeatability**
With positional accuracy of ± 0.08 mm and repeatability within ± 0.12 mm, the plotter outperforms hand-drawing by an order of magnitude, making it suitable for technical diagrams, PCB layouts, and light engraving.
- **Desktop-Friendly Footprint**
At $300 \times 300 \times 200$ mm, the machine fits standard school desks, home workbenches, and confined workshop benches, enabling on-demand operation without dedicated floor space.
- **Open-Source & Localized Design**
Leveraging CATIA V5, Cura slicing, Arduino + GRBL, and free vector tools ensures no recurring licenses and fosters a DIY ecosystem across Indian colleges, Atal Tinkering Labs, and grassroots makerspaces.

8.2 Real-World Impact

- **Empowering** **Education**
Students gain hands-on experience with G-code, CAD-to-fabrication workflows, and additive manufacturing principles, reinforcing STEM curricula with project-based learning.
- **Catalyzing** **Prototyping**
Hobbyists and startups can iterate designs in minutes—whether plotting circuit masks or sketching product mockups—without costly outsourcing or large CNC infrastructure.
- **Supporting** **MSMEs**
Small-batch label making, stencil cutting, and light engraving can be internalized, reducing lead times from days to minutes and per-job costs by over 80%.

By sourcing over 80% of the structure through local 3D printing and off-the-shelf electronics, the project exemplifies self-reliant manufacturing and technology democratization.

8.3 Strategic Insights

1. Iterative Prototyping Wins

Rapid CAD–print–test cycles in CATIA V5 and Cura refined both mechanical rigidity and assembly ease, underscoring the value of additive fabrication for functional hardware.

2. Balance of Simplicity & Performance

An open-loop GRBL control system delivered robust drawing performance; future closed-loop upgrades can elevate the platform to light-milling and high-speed engraving.

3. Community Drives Innovation

Open documentation and a modular design invite contribution—from custom toolheads to UI enhancements—fueling a shared innovation roadmap that adapts to diverse user needs.

8.4 Path Forward

While the current prototype fulfils its mandate, several enhancements can amplify its utility:

- Integrate closed-loop feedback with encoders for zero-error motion control.
- Upgrade to a stepper-driven **Z-axis** for controlled-depth engraving and light milling.
- Develop a **modular** multi-tool carriage to seamlessly switch between pens, lasers, and drag knives.
- Add an onboard touchscreen and wireless connectivity for standalone operation and cloud-based job management.

Our Mini CNC Plotter stands as a tangible proof of concept that precision automation can be both accessible and locally sustainable. By championing open-source principles and leveraging India’s burgeoning 3D-printing capabilities, we’ve laid a foundation for widespread adoption—from classrooms to micro-factories—paving the way for a new generation of makers and innovators. The Mini CNC Plotter Machine developed through this project represents a significant step toward making precision automation accessible, affordable, and locally manufacturable in India. By leveraging 3D-printed components, open-source electronics, and freely available software tools, we successfully created a compact plotting device capable of producing accurate 2D sketches, circuit layouts, and educational diagrams.



The project addressed a real-world gap—bridging the divide between expensive industrial CNCs and imprecise manual methods. With a build cost under ₹6,000 and repeatability of ± 0.1 mm, the plotter proved effective for STEM education, maker experimentation, and small-scale industrial tasks such as label making and light engraving.

Beyond its immediate functionality, this machine serves as a teaching aid, a prototyping companion, and a gateway into digital manufacturing. It encourages iterative design, introduces G-code programming concepts, and showcases the real-world benefits of additive manufacturing.

By successfully integrating mechanical design, electronics, and software control, this project exemplifies the spirit of innovation and self-reliance promoted by India's "Make in India" and "Digital India" movements. It lays a foundation for future improvements such as tool-head modularity, enhanced firmware, and IoT integration—ensuring the plotter remains adaptable to the evolving needs of learners, innovators, and industry professionals alike.

9. References

The design and development of our Mini CNC Plotter Machine were informed by multiple open-source and community resources:

- The core project concept and step-by-step build sequence were inspired by the YouTube tutorial “Mini CNC Plotter Machine” (https://youtu.be/yjwiEk_ZV5E).
- Mechanical design principles and parametric modeling techniques were guided by CATIA V5 user documentation from Dassault Systèmes.
- Print-preparation workflows followed Ultimaker Cura slicing guidelines and best practices for PLA from Ultimaker’s online manual.
- Firmware integration leveraged the open-source GRBL v1.1 specification and Arduino Uno reference materials.
- Vector-to-G-code conversion methods were drawn from Inkscape and JSCut community tutorials.
- Component specifications (NEMA 17 steppers, A4988/DRV8825 drivers, GT2 belts, LM8UU bearings, SG90 servos) were sourced from manufacturer datasheets and distributors such as <https://amzn.in/d/1vcyqk4>.
- **Firmware and Source Code**
GRBL 28b – [GitHub Repository](#)
Used for motion control and G-code interpretation via Arduino Uno.
- **3D-Printed Components**
Thingiverse Model – [3D Printed Framework](#)
Adapted mechanical parts for the plotter structure and pen carriage.
- **Vector Design Software**
Inkscape v1.2 – [Download Link](#)
Used to create and export SVG files for technical drawings.
- **Online G-code Generator**
JSCut – [Web Tool](#)
Converted SVG files into CNC-compatible G-code.



- **G-code Sender**

Universal G-code Sender (UGS) – [UGS Platform](#)

Used to communicate between PC and CNC hardware for job execution.

We gratefully acknowledge these freely available tools, tutorials, and datasheets, which made it possible to prototype an affordable, locally replicable CNC plotter for Indian educational and small-scale industrial use.